2D Metal Organic Frameworks

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March 10th, 2020

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Introduction

Over the past 20 years, metal-organic frameworks (MOFs) have been extensively studied for their healthcare, environmental, and technological applications such as gas separations, energy storage and conversion, solar and fuel cells, capacitors and catalysis. MOFs are composed of inorganic metallic nodes and organic linkers, or ligands, which undergo molecular self-assembly into atomically thin layered two-dimensional structures held together through intermolecular forces. Thus, MOFs are structurally rigid with permanent porosity which results in a high surface-to-volume ratio and high chemical adsorption capacity. The continued research attempts to obtain both macro and nano-crystalline MOFs of various shapes and sizes by altering composition through linker design and post-synthetic modification. MOFs functionalized with amino groups have been found to form nanotubes with the amino groups within the channels. Research has shown that incorporating second functionalized groups such as Cl, Br, or OH groups enhance the electrochemical properties of the MOFs. These secondary functional groups enhance the adsorption and desorption of charged ions. The desired functionality of a MOF can be obtained by selecting the appropriate ligands. However, MOFs geometry is determined by the steric bulk of the linker molecule. Currently, the most conductive MOFs are those with a planar, graphene-like structure where the charge carriers are confined to the 2D lattice. Metal centers with unpaired electrons and use of organic linkers with redox active molecules or stable radicals creates a high charge delocalization.

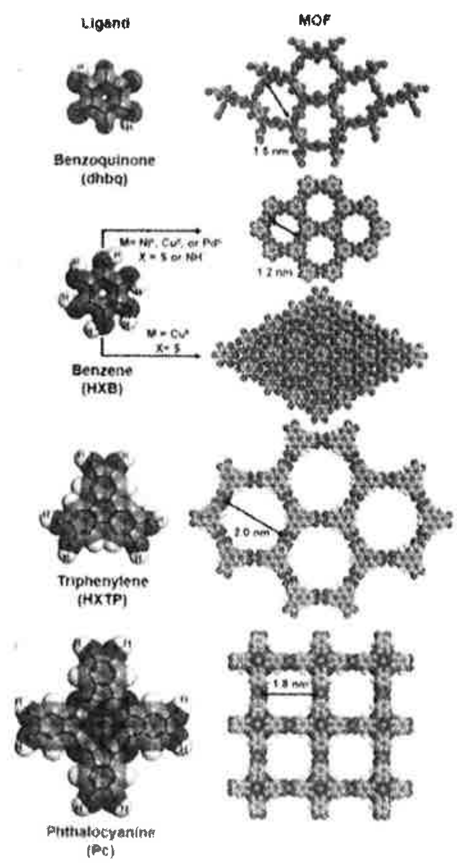

Figure 1 (Ko, Mendecki, & Mirica, 2018)**:** Novel MOFs and their single units. Showing the difference in linkers leading to different geometries.

Methods

MOF synthesis under solvothermal conditions is the most common method. In general, metal salt hydrate complexes are combined with the organic linkers and solvents composed of the intended linkers. This mixture is placed into an autoclave and into an oven for 24+ hours. The crystalline product is then obtained by vacuum filtration and rinsed. Once dry, crystals are analyzed using x-ray diffraction (XRD) and a scanning electron microscope (SEM). Electrochemical properties are determined using a standard three-electrode test pool and electrochemical impedance spectroscopy.

Results

The crystalline products are frequently in low yield and are particularly small, therefore XRD is commonly done using pellets and polycrystalline films. XRD and SEM confirm the MOFs synthesized are comprised of a 2D layered network, contained 1D channels. Electrode tests show that the capacitance is close to the ideal rectangular shape and consistent with the electric double-layer capacitance. Conductivity is commonly plotted against temperature

Challenges

The major change is realistically obtaining the designed MOF with the desired topology and functionality. Synthesizing crystals which solely grow along the vertical direction without affecting the two other lateral directions.

Furthermore, MOFs inherently have low electrical conductivity which must be enhanced through better design to increase the concentration and mobility of charge carriers. Increasing charge carriers by thermal or photo-excitation, doping, and hole or electron injection. Increasing charge mobility by minimizing grain boundaries and reducing the number of charge trapping sites.

Future Work

Future work should focus on the development of MOFs with higher conductivity through linker design and post-synthetic modification, such as doping. Post-synthetic modification incorporates new molecules to the MOFs that act as bridges between neighboring MOF units which promote charge transport. However, doping can lead to a decrease in surface area, porosity, pore volume, and stability. MOFs development should limit doping and instead look to redox active organic linkers which have little affect on structure and substantially increase electrical conductivity.

Metal-organic framework nanosheets (MOFNs) have a higher flexibility which allows for use in portable devices and nanotechnology applications. These 2D nanosheets have high amounts of exposed surface atoms and active sites which allows for short paths for mass transport which increases molecular movement and separation flux.

References

Reference Citations in Text ?

Day, R. W., Bediako, D. K., Rezaee, M., Parent, L. R., Skorupskii, G., Arguilla, M. Q., . . . Dinca, M. (2019). Single Crystals of Electrically Conductive Two-Dimensional Metal-Organic Frameworks: Structural and Electrical Transport Properties. *ACS Central Science*(5), 1959-1964.

Dhakshinamoorthy, A., Asiri, A. M., & Garcia, H. (2019). 2D Metal-Organic Frameworks as Multifunctional Materials in Heterogeneous Catalysis and Electro/Photocatalysis. *Advanced Materials*(31).

Duan, J., Li, Y., Pan, Y., Behera, N., & Jin, W. (2019). Metal-Organic Framework Nanosheets: An Emerging Family of Multifunctional 2D Materials. *Journal of Coordination Chemistry Reviews*(395), 25-45.

Ko, M., Mendecki, L., & Mirica, K. A. (2018). Conductive Two-Dimensional Metal-Organic Frameworks as Mulifunctional Materials. *ChemComm*(54), 7873-7891.

Liu, X., Valentine, H. L., Pan, W.-P., Cao, Y., & Yan, B. (2016). 2D Metal-Organic Frameworks: Syntheses, Structures, and Electrochemical Properties. *Journal of Inorganica Chimica Acta*(447), 162-167.